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GENERAL INTRODUCTION

Organizational practices are now dependent not only on human resources but also on the ability to manage processes effectively and adapt to rapid change driven by the evolution of digital technologies. In this context, recruiting is strategically important, as it directly impacts team building and organizational success. However, the complexity of this process has increased because of the growing number of applications and the demand for faster, more accurate treatment. Consequently, organizations are turning to digital solutions that centralize information, improve workflows, and support decision-making, turning recruitment into a structured process that requires efficient tools.

In this setting, Sopra HR Software is recognized as a leading player in human resource management solutions. Operating in a field where technological innovation and service quality are essential, the company emphasizes the need to improve its internal processes and business tools. In this sphere, the digitalization of recruitment is a major lever for streamlining application processing, enhancing the user experience, and supporting recruiters in their day-to-day activities.

The implementation of an intelligent recruitment portal aims to meet both operational and strategic needs by providing a more cohesive framework for managing job offers, applications, and interactions among all stakeholders involved in recruitment. Consequently, a central question arises : how can a recruitment portal be designed to centralize applications, facilitate CV processing, and effectively support recruiters in their decision-making, while aligning with a logic of continuous improvement in the recruitment process ?

To explore the identified problem and present the work conducted during the internship, the report is divided into three main chapters :

- **Chapter One** presents the general framework of the project by introducing the host organization and situating the project within its context. It analyzes the existing recruitment process and its limitations, compares recruitment solutions, and justifies the proposed intelligent recruitment portal.
- **Chapter Two** addresses requirements analysis and system design. It describes the functional and non-functional requirements, highlights the expected features, and introduces the main design components used to build the portal.

- **Chapter Three** is devoted to the development and implementation phase. It outlines the steps involved in building the application, details the implemented features, and summarizes the main outcomes achieved.

Finally, the report concludes with a summary that recaps the context of the work, reviews the project, and presents possible directions for future enhancements.

Chapter

1

GENERAL PROJECT CONTEXT

Plan

Introduction

This chapter establishes the project's general framework by introducing the host organization and situating it within its institutional and operational context. It examines the existing recruitment process at Sopra HR Software and highlights its limitations through detailed analysis. A comparative study of available recruitment solutions is then presented to identify gaps in current approaches. Finally, the chapter justifies the proposed intelligent recruitment portal as a response to the identified needs and challenges.

1.1 Presentation of the Host Organization

Sopra Steria is a well-known European group recognized for its expertise in consulting, digital services, and business software publishing, particularly in human resources and banking. In 2024, the Sopra Steria Group operates in nearly 30 countries and employs around 51,000 people. Sopra HR Software is a subsidiary of the Sopra Steria Group, specializing in Human Resources management, payroll administration, and talent management, both nationally and internationally. More than 850 clients use its solutions across over 54 countries. Designed to meet the specific needs of HR departments, these solutions are intended for large and medium-sized enterprises in both the public and private sectors.

FIGURE 1.1 – Logo of Sopra HR Software

1.2 Project Study

This section outlines the project context and analyzes the existing system to identify its limitations. It also presents a comparative study of current recruitment portals, highlighting their strengths and shortcomings relative to the proposed platform. This analysis supports the definition and justification of the proposed solution.

1.2.1 Project Context

This project was carried out as part of a Final Year Project at the Higher Institute of Technological Studies of Nabeul, to apply academic knowledge to the development of an innovative software solution. It focuses on the design and implementation of an intelligent recruitment portal intended to support and modernize the recruitment processes of Sopra HR Software. In today's digital environment, recruitment platforms play a key role in streamlining hiring processes and improving interaction between candidates and recruiters. Accordingly, this project aims to design a centralized and intelligent recruitment platform tailored to the needs of Sopra HR Software, leveraging modern technologies to enhance efficiency and user experience.

1.2.2 Analysis of the Existing System

The recruitment process in many organizations, including Sopra HR Software, relies on manual procedures and multiple disconnected tools, where job offers and applications are managed across various platforms. As a result, recruiters must manually handle tasks such as CV screening, shortlisting, and interview coordination. As highlighted in studies on online recruitment systems, handling large volumes of applications without an automated applicant tracking system makes it difficult to identify suitable candidates efficiently and consistently. This fragmentation slows down the recruitment process and increases the workload on HR teams.

Current Recruitment Practices at Sopra HR Software

Several limitations characterize the existing recruitment approach :

- **Lack of centralization** : Candidate data and job offers are spread across multiple platforms.
- **Time-consuming CV screening** : Manual sorting of resumes increases processing time and recruiter workload.
- **Reduced decision accuracy** : Comparing candidates becomes difficult without structured evaluation tools.
- **Limited process visibility** : Recruiters struggle to track application status and recruitment stages effectively.

These limitations reduce the overall efficiency of recruitment activities and prevent recruiters from focusing on strategic HR tasks.

1.2.3 Comparative Study of Existing Recruitment Solutions

This comparative study analyzes existing recruitment platforms as reference systems to identify best practices and functional limitations. Its objective is to highlight the gaps between generic recruitment solutions and the specific needs of an internal recruitment portal dedicated to Sopra HR Software Tunisia.

Aspect	Public Recruitment Platforms (LinkedIn, Indeed, etc.)	ATS-Oriented Tools (Workable)	Proposed Internal Portal (Sopra Recruit)
Target Scope	Multi-Company, Global	Multi-Company	Single Company (Sopra HR Software)
Recruitment Context	Generic	Semi-Customizable	Fully context-specific
Job Offer Management	Public and standardized	Centralized	Dedicated to internal positions
Candidate Flow	High volume, external candidates	Controlled pipelines	Targeted candidates
Customization	Limited	Moderate	High
Adaptation to Local Process	Low	Moderate	Complete
Data Ownership	Platform-dependent	Platform-dependent	Fully internal
Recruitment Monitoring	Standard statistics	Advanced pipelines	Tailored dashboards
Main Limitations	Lack of customization	Cost and rigidity	Scope limited to one company

TABLE 1.1 – **Table 1.1 : Comparative Overview of Recruitment Solutions**

The comparison shows that public recruitment platforms are primarily generic and insufficiently adapted to company-specific processes, while ATS tools remain constrained by predefined workflows. In contrast, the proposed internal portal is tailored to Sopra HR Software, enabling better integration, customization, and data control.

1.2.4 Proposed Solution

To address these limitations, this project proposes the design and development of an intelligent recruitment portal for Sopra HR Software. The portal aims to digitalize and automate the recruitment process by integrating the following features :

- Centralized management of job offers and applications
- Automated CV collection and structured candidate profiles
- Support for CV screening and candidate shortlisting
- Tools to assist recruiters in evaluating candidates and making decisions
- A unified platform facilitating interaction between recruiters and applicants

By adopting an intelligent recruitment portal, Sopra HR Software can streamline hiring workflows, reduce recruitment time, and enhance the effectiveness and reliability of the recruitment process. This solution aligns with best practices in electronic recruitment systems, which have proven effective in simplifying hiring and improving organizational performance.

1.3 Working Methodology Choice

During the development of an information system, it is essential to adopt a clear, structured methodology to guide the project through its different stages. A development methodology defines the processes and steps required to organize the work, ensure progress at each phase, and align the system being developed with user needs. For this project, the Agile methodology was chosen, as it provides flexibility, encourages continuous improvement, and supports incremental delivery throughout the development cycle.

1.3.1 Agile Method

Agile methodology is widely adopted in software development and IT service companies due to its ability to manage complexity and continuous change. According to the Agile Manifesto, Agile is based on an iterative and incremental development approach that emphasizes frequent delivery of functional software, close collaboration between stakeholders, and adaptability to changing requirements. The manifesto highlights values such as individuals and interactions over processes and tools, and responding to change over following a plan.

In contrast to traditional predictive models, Agile encourages delivering software in small, usable increments, allowing stakeholders to provide continuous feedback and enabling teams to adjust the system progressively. As stated by Atlassian, Agile divides development work into short cycles, improving responsiveness, transparency, and product quality through constant evaluation and refinement.

Among the various Agile frameworks, SCRUM was chosen for this project and is discussed in the next section.

1.3.2 SCRUM Methodology

Scrum is a lightweight Agile framework widely used to manage and develop complex systems in an iterative, incremental manner. According to the Scrum Guide, Scrum structures development activities into fixed-length iterations called sprints, which typically last from one to four weeks. Each sprint is executed with a clear objective and results in the delivery of a potentially usable product increment, allowing continuous inspection, adaptation, and value creation throughout the project lifecycle.

Scrum applies Agile principles by emphasizing collaboration, frequent delivery, and responsiveness to change. It is founded on empiricism, which relies on transparency, inspection, and adaptation as core pillars, enabling teams to progressively refine the product while responding effectively to evolving requirements.

1.3.3 SCRUM Actors

The Scrum framework defines three key roles, each with specific responsibilities :

- **Product Owner** : Responsible for defining the product vision and representing stakeholder needs. The Product Owner manages and prioritizes the product backlog to maximize the value delivered by the development team.
- **Scrum Master** : Ensures that the Scrum framework is properly understood and applied. The Scrum Master facilitates communication, removes impediments, and promotes adherence to Agile principles.
- **Development Team** : A self-organized and cross-functional team, typically composed of 3 to 9 members, responsible for designing, developing, and delivering functional increments at the end of each sprint.

The structured yet flexible framework enables effective collaboration, continuous improvement, and high-quality delivery throughout the development process.

Chapter

2

REQUIREMENTS ANALYSIS AND SPECIFICATION

Plan

Introduction

This chapter is dedicated to the project's preparation phase. It begins with the elicitation and specification of system requirements, then presents the functional and non-functional requirements of the system. The chapter also describes the organization and management of the project using the SCRUM methodology, including team roles, the product backlog, and sprint planning. It then presents the working environment in terms of hardware, tools, and technologies used during development, and concludes with the requirements modeling through a global use case diagram.

2.1 Requirements Elicitation

The purpose of requirements elicitation is to identify the stakeholders' needs and expectations in the recruitment process. In this project, requirements were gathered through an analysis of the existing recruitment system at Sopra HR Software and discussions with recruiters to understand their difficulties and daily practices. This step highlighted the need for a centralized recruitment platform, better candidate tracking, and improved support for recruiter decision-making, which serve as the basis for the formal requirements defined in the next section.

2.1.1 Requirements Specifications

The specification of requirements begins during the preliminary analysis phase, which marks the start of the system development process. This phase focuses on identifying and formalizing both functional and non-functional requirements of the proposed recruitment portal, based on the information gathered during requirements elicitation. These requirements are expressed using clear textual descriptions and simple models to ensure a precise understanding of the system's expected behavior and constraints.

2.1.2 Identification of Actors

An information system interacts with different users, each having specific roles and responsibilities. In the proposed recruitment portal, these users are identified as system actors and are presented in Table 2.1.

Actor	Description
Administrator / HR	Manages the recruitment platform by creating job offers, configuring scoring criteria, and supervising applications and system settings.
Manager	Reviews applications related to their department and contributes to candidate evaluation and selection.
Candidate	Applies for job offers by submitting a CV and tracks the status of submitted applications.
Visitor	Browses job offers without authentication and registers to become a candidate.

TABLE 2.1 – Identification of System Actors

2.2 Functional and Non-Functional Requirements

In this section, we present the functional and non-functional requirements for the development of our portal, Sopra Recruit. The proposed solution aims to centralize the management of job offers and applications, automate CV analysis through parsing and scoring mechanisms, and support recruiters in candidate pre-selection and decision-making. These requirements define the expected behavior of the system as well as the quality constraints necessary to ensure efficiency, reliability, and security.

2.2.1 Functional Requirements

The functional requirements of the intelligent recruitment portal are organized according to the different user profiles interacting with the system. These profiles include the Visitor, the Candidate, the Human Resources Officer, the Manager, and the automated system. Each actor is associated with a specific set of functionalities that define their interactions with the platform.

Visitor (Unauthenticated User)

The Visitor represents a user who accesses the platform without authentication. The functional needs associated with this actor are as follows :

- Browse the institutional pages of the recruitment portal.
- Consult the list of published job offers.

- Search and filter job offers according to defined criteria.
- View detailed information about a selected job offer.
- Create a candidate account through a public registration form.
- Activate the account by confirming the email address.

Candidate

The Candidate is an authenticated user who applies for job offers and manages personal applications. The Candidate's functional requirements are :

- Sign in and sign out of the platform securely.
- Manage personal account information and update profile data.
- Consult available job offers and their detailed descriptions.
- Apply to job offers by submitting a CV and complementary information.
- Reuse an existing CV or upload a new one during application submission.
- Track the status of submitted applications.
- Consult planned interviews and receive related notifications.

Human Resources Officer (HR) / Administrator

The Human Resources Officer acts as the administrator of the recruitment platform and is responsible for managing and supervising the entire recruitment process. This role has full access to the system and oversees both functional and organizational aspects of the application. The functional requirements for this role include :

- Access the secured administration interface.
- Manage user accounts and assign roles and permissions.
- Create, publish, update and archive job offers.
- Define and analyze received applications.
- Evaluate applications and decide on rejection or progression.
- Forward selected applications to managers.
- Plan and manage recruitment interviews.
- Review and modify job offers proposed by managers, and decide whether to approve or reject them, with automatic publication upon approval.
- Monitor recruitment activity through dashboards and reports.

Manager

The Manager is an authenticated internal user who participates in the recruitment decision-making process within a specific department. The platform may involve multiple managers, each responsible for evaluating candidates and proposing recruitment needs related to their organizational unit. The functional requirements associated with this actor include :

- Access the secured management interface.
- Consult the applications forwarded by the HR department.
- Review candidate profiles and evaluation results.
- Validate or reject candidate applications.
- Participate in interview follow-up.
- Manage personal notifications and profile information.
- Propose a new job offer by submitting a dedicated request form when a position becomes vacant or requires reinforcement.
- Submit the proposed job offer to the Human Resources Officer for review and validation.

2.2.2 Non-Functional Requirements

Beyond the functionalities described above, the recruitment portal must meet a set of quality requirements that determine its acceptance by end users (candidates, HR officers, managers) and the platform's long-term sustainability. These requirements are formulated as observable constraints rather than implementation choices.

Security and Confidentiality

- Users must be logged in to access protected parts of the system.
- Users can only perform actions allowed by their role (Candidate, HR, or Manager).
- Passwords and sensitive personal data (such as CVs and contact details) must be stored securely.
- A candidate account is activated only after the email address is confirmed within a limited time.
- All communication between front-end, back-end, and AI service must be secure.
- The platform must respect data-protection rules, including user consent, access to personal data, and controlled storage of CVs.

Performance and Responsiveness

- The main pages (dashboards, application lists, calendars) should remain smooth and responsive, with loading times under 2 seconds under normal use.
- CV analysis and scoring should be completed quickly, with a target processing time of less than 10 seconds per application.
- AI processing should run in the background so that candidates can submit their applications without waiting.

Reliability and Availability

- The platform should be available at least 99% of the time during business hours.
- If the AI service is temporarily unavailable, applications must still be saved, and scoring can be recalculated later.
- The recruitment process must remain consistent, and important errors must be recorded instead of being ignored.

Usability and Ergonomics

- The interface should have a modern and professional design, and each screen should follow the natural way users work.
- Every action performed by a user should immediately show visual feedback, and the system status should be displayed consistently across the application.
- The application process should guide candidates step by step and include a clear GDPR consent step.

Compatibility and Accessibility

- The platform should be accessible through major web browsers (Chrome, Edge, Firefox, Safari) without requiring any installation.
- The interface should work well on desktop, laptop, and tablet screens, use consistent language, and allow adding other languages in the future.

Scalability

- The platform should support a growing number of users, job offers, and applications without noticeable slowdowns, and its main features should remain independent.

Maintainability and Evolvability

- The source code should be well-organized into clear modules and documented so that a new developer can easily understand and maintain the project.
- Configuration settings (database, AI service, email server, security keys) should be kept separate from the code so the application can run in different environments.
- HR users should be able to adjust the recruitment workflow and scoring criteria without needing a new software update.

Portability and Deployment

- The platform's components should be deployable separately, for example, on cloud or container-based systems, without changing the application logic.

2.3 Project Management Using SCRUM

To manage and develop the intelligent recruitment platform, the Agile SCRUM approach was adopted. SCRUM is highly suitable for this project because of its iterative nature and flexibility in handling evolving requirements. This approach allows continuous improvement of the product while ensuring regular delivery of functional increments.

The SCRUM methodology is based on the following principles :

- Continuous collaboration between team members and stakeholders to ensure a shared understanding of project objectives.
- Regular ceremonies such as Sprint Planning, Daily Meetings, Sprint Reviews, and Sprint Retrospectives, which help structure the work and monitor progress.
- Iterative development through sprints, which are short development cycles with clearly defined objectives.
- Constant monitoring of project progress, enabling early detection of issues and gradual delivery of a stable and high-quality product.

This approach allowed the team to prioritize essential functionalities, integrate feedback progressively, and adapt the platform to real user needs throughout the development process.

2.3.1 Team and Roles

The Product Owner (PO)

The Product Owner is responsible for defining the functional requirements of the project and ensuring that the product delivers value to its users. The Product Owner's responsibilities include :

- Defining and prioritizing project needs based on user expectations and business objectives.
- Managing and maintaining the product backlog.
- Validating completed functionalities at the end of each sprint.
- Collaborating closely with stakeholders to ensure their requirements are properly understood and implemented.

In this project, the Product Owner role was fulfilled by Mr. Ramzi Ben Chamekh, Manager at Sopra HR Software.

SCRUM Master (SM)

The Scrum Master ensures that SCRUM principles and practices are properly applied throughout the project. This role focuses on facilitating collaboration within the team and supporting the development process by removing obstacles. The Scrum Master's responsibilities include :

- Ensuring correct application of the SCRUM framework.
- Facilitating communication between team members.
- Assisting the team in resolving impediments.
- Coordinating SCRUM ceremonies and supporting sprint execution.

In this project, the Scrum Master role was assumed by Ms. Nada Bouhlel, the project supervisor.

Development Team

The Development Team is responsible for the concrete implementation of the intelligent recruitment platform. Team members collaborate actively during each sprint to design, develop, and test the system's functionalities. The responsibilities of the development team include :

- Designing and implementing platform features.

- Participating in sprint planning, daily meetings, and reviews.
- Delivering functional increments at the end of each sprint.

In this project, the development team consists of Mayssem ESSMAII and Bechir JABLI.

2.3.2 Product Backlog

The product backlog constitutes the ordered list of all features and requirements to be implemented in the intelligent recruitment portal. Each item is expressed as a user story and is assigned a priority level and a complexity estimate using story points. The backlog is organized into thirteen functional epics covering all major system capabilities.

Epic	Actor	ID	User Story	Priority	Complexity
Epic 1 Authentication & Registration	Visitor	US-AUTH-01	As a visitor, I want to create an account with my first name, last name, email and password, so that I can access the platform as a candidate.	High	3
	Candidate	US-AUTH-02	As a candidate, I want to receive an activation email and verify my account by clicking the link, so that my identity is confirmed before I can sign in for the first time.	High	2
	Candidate / Admin RH / Manager	US-AUTH-03	As an authenticated user, I want to sign in with my email and password, so that I can access the functionalities corresponding to my role.	High	2
	Candidate / Admin RH / Manager	US-AUTH-04	As an authenticated user, I want to sign out of the platform, so that my session is closed securely.	High	1
Epic 2 User Management	Admin RH	US-USR-01	As an Admin RH, I want to create user accounts (Candidate, Manager, Admin RH) directly as active, so that new members can access the platform immediately without going through email verification.	High	2
	Admin RH	US-USR-02	As an Admin RH, I want to list and update user information, so that user records remain accurate and up to date.	High	2
	Admin RH	US-USR-03	As an Admin RH, I want to assign or change a user's role, so that each user accesses only the functionalities relevant to their responsibility.	High	3

Epic	Actor	ID	User Story	Priority	Complexity
Epic 3 User Profile	Candidate / Admin RH / Manager	US-PROF-01	As an authenticated user, I want to view and update my personal profile (phone number, address, profile picture), so that my information remains accurate and my identity is recognisable on the platform.	Medium	3
Epic 4 Job Offer Management	Admin RH	US-OFF-01	As an Admin RH, I want to create, update, change the status (draft, published, archived) and delete job offers, so that the list of available positions is always accurate and up to date.	High	5
	Admin RH	US-OFF-02	As an Admin RH, I want to consult all offers regardless of their status, so that I have a complete and permanent view of the recruitment pipeline.	High	2
	Visitor / Candidate	US-OFF-03	As a visitor or candidate, I want to browse published job offers and filter them by keyword, experience level, education level and contract type, so that I can quickly identify relevant opportunities.	High	3
Epic 5 Application Management	Candidate	US-CAND-01	As a candidate, I want to apply to a job offer by submitting my CV, a cover letter and my GDPR consent through a guided wizard, so that my application is formally and completely registered in the system.	High	5
	Candidate	US-CAND-02	As a candidate, I want to consult the list and the details of my applications with their current status, so that I am always informed about the progress of my candidacy.	High	2
	Admin RH	US-CAND-03	As an Admin RH, I want to consult, filter and sort all received applications by status, score or date, so that I can efficiently manage the recruitment pipeline.	High	3
	Admin RH	US-CAND-04	As an Admin RH, I want to process an application by placing it under evaluation, forwarding it to a chosen manager, or rejecting it, so that only qualified profiles progress in the recruitment workflow.	High	3

Epic	Actor	ID	User Story	Priority	Complexity
	Manager	US-CAND-05	As a Manager, I want to consult the applications forwarded to me and validate or reject them, so that I exercise my responsibility in the final selection of candidates.	High	3
	Admin RH / Manager	US-CAND-06	As an Admin RH or Manager, I want to download the original CV file of a candidate, so that I can review the source document directly.	High	2
Epic 6 CV & AI Analysis	Admin RH	US-CV-01	As an Admin RH, I want to consult the structured data automatically extracted from a candidate's CV (skills, experiences, education, languages, certifications, soft skills), so that I can review the candidate's profile without manually reading the raw document.	High	5
Epic 7 AI Scoring	Admin RH	US-SCORE-01	As an Admin RH, I want to define, update and delete weighted scoring criteria (skills, experience, education, languages, certifications) per job offer, so that candidates are evaluated objectively against the specific requirements of each position.	High	5
	Admin RH	US-SCORE-02	As an Admin RH, I want to consult the AI matching score and its breakdown per criterion for each application, so that I can make objective and informed selection decisions.	High	3
	Admin RH	US-SCORE-03	As an Admin RH, I want to sort applications by descending score and manually trigger a score recalculation when needed, so that the candidate ranking always reflects the latest criteria.	Medium	2
Epic 8 Interviews & Calendar	Admin RH	US-ENT-01	As an Admin RH, I want to schedule an interview for a validated application by specifying the date, location and format (in-person, video or phone), so that the recruitment process advances in a structured and traceable manner.	High	5

Epic	Actor	ID	User Story	Priority	Complexity
	Admin RH	US-ENT-02	As an Admin RH, I want to confirm or cancel a planned interview, so that the calendar remains accurate and all parties are informed.	Medium	2
	Admin RH / Manager	US-ENT-03	As an Admin RH or Manager, I want to consult the interview calendar relevant to my role, so that scheduling conflicts are avoided and workload is managed effectively.	High	3
	Candidate	US-ENT-04	As a candidate, I want to consult my upcoming interviews with their date, location and format, so that I can prepare accordingly.	High	2
Epic 9 Evaluations	Admin RH	US-EVAL-01	As an Admin RH, I want to evaluate an application with a structured comment and decision (to follow up, to review, to reject), so that my assessment is formally recorded and traceable.	High	3
	Manager	US-EVAL-02	As a Manager, I want to evaluate a forwarded application with a comment and a decision, so that my validation is documented and directly influences the status of the application.	High	3
	Admin RH / Manager	US-EVAL-03	As an Admin RH or Manager, I want to consult the full evaluation history of an application, so that all assessments are visible and the decision trail is traceable.	Medium	2
Epic 10 Offer Recommendation	Manager	US-REC-01	As a Manager, I want to draft and submit a job offer recommendation to the Admin RH when a position becomes available, so that recruitment needs are formalised and communicated promptly.	Medium	3
	Admin RH	US-REC-02	As an Admin RH, I want to consult, edit, accept or reject an offer recommended by a Manager, so that I retain full control over the offers published on the platform.	Medium	3

Epic	Actor	ID	User Story	Priority	Complexity
Epic 11 Notifications	Candidate / Admin RH / Manager	US-NOTIF-01	As an authenticated user, I want to receive notifications for key recruitment events (application submitted, status changed, interview scheduled, account activated) and mark them as read individually or all at once, so that I remain informed without having to check the platform manually.	High	3
Epic 12 Dashboard & Statistics	Admin RH	US-DASH-01	As an Admin RH, I want to consult a recruitment dashboard displaying key indicators (total applications, published offers, selection rate, average score, status distribution), so that I can monitor and steer recruitment activity in real time.	High	5
	Admin RH	US-DASH-02	As an Admin RH, I want to consult detailed statistics (applications per month, top offers, average recruitment time) and export a statistical report, so that I can analyse recruitment performance and share results with management.	Medium	5
Epic 13 Public Site	Visitor	US-PUB-01	As a visitor, I want to browse the landing page and institutional pages (About us, Our professions, Our sectors, Why choose us, Culture and presence), so that I can learn about Sopra HR Software and its opportunities before registering.	Medium	3

TABLE 2.2 – Product Backlog — Intelligent Recruitment Portal (Sopra HR)

2.3.3 Sprint Planning

After the definition of the product backlog, the workload is distributed across the planned sprints. This distribution is presented in Table 2.3, which outlines the main activities and objectives of each sprint.

Sprint	Main Tasks	Duration
Sprint 1 — Analysis and Setup	Requirements analysis and system design ; project architecture setup ; authentication and user management development ; backend, frontend, and database configuration.	4 weeks
Sprint 2 — Recruitment Management	Job offer management ; application and CV management ; HR officer and candidate interfaces ; search, filtering, and application tracking.	4 weeks
Sprint 3 — Artificial Intelligence	CV extraction and structured data analysis ; CV-to-offer matching system development ; scoring engine implementation ; scoring criteria configuration and AI-based recommendations.	4 weeks
Sprint 4 — Finalization and Deployment	Interview scheduling and candidate evaluation management ; HR dashboard and recruitment statistics ; testing, optimization, and bug fixing ; deployment and project documentation.	4 weeks

TABLE 2.3 – Sprint Planning

2.4 Working Environment

This section describes the working environment used throughout the development of the intelligent recruitment portal. It covers the hardware configuration, the software tools, and the technologies selected to implement the various components of the system.

2.4.1 Hardware Environment

Component	Specification
Model	HP EliteBook 840 G8
RAM	16GB
CPU	11th Gen Intel Core i5-1145G7 2.60 GHz
GPU	Intel UHD Graphics
Operating system	Windows 11

TABLE 2.4 – Hardware Environment

2.4.2 Software Environment

Operating System : Windows 11 is the operating system installed on the computer used to build this project. An operating system is the main software that manages the computer and allows other programs to run on it. Windows 11 offers a clean and modern working space that works well with many development tools, making it easy to write, test, and run the application without any problems.

FIGURE 2.1 – Windows 11 Logo

2.4.3 Development Tools

Visual Studio Code : Visual Studio Code, also called VS Code, is a free and lightweight code editor made by Microsoft. It was used to write and edit the frontend code of this project. VS Code is very popular among developers because it is simple to use, starts up quickly, and supports many programming languages. It also has a large collection of extensions, which are small add-ons that add useful features like code suggestions, error detection, and tools for working with Git.

FIGURE 2.2 – Visual Studio Code Logo

IntelliJ IDEA : IntelliJ IDEA is a professional code editor made by JetBrains, designed mainly for writing Java applications. It was used to build the backend of this project. It helps developers write code faster by offering smart suggestions, automatic error detection, and built-in tools for running and testing the application. It is widely used in the industry for building large and reliable backend systems.

FIGURE 2.3 – IntelliJ IDEA Logo

Overleaf : Overleaf is a free online tool used to write documents using LaTeX, which is a system that formats text in a professional and structured way. It was used to write this project report. Overleaf is easy to use because it works directly in the browser without needing to install anything on the computer, and it shows a live preview of the document while editing.

FIGURE 2.4 – Overleaf Logo

Git : Git is a free tool that helps developers keep track of all the changes made to the code over time. It is called a version control system. With Git, it is possible to save different versions of the project, go back to an older version if something goes wrong, and work with a team without accidentally overwriting each other's work.

FIGURE 2.5 – Git Logo

StarUML : StarUML is a diagram drawing tool used to design and show the structure of a software system. In this project, it was used to draw UML diagrams, which are special diagrams that show how different parts of the application are connected and how they work together. These diagrams make it easier to understand and explain the system before writing any code.

FIGURE 2.6 – StarUML Logo

Swagger : Swagger is a tool used to document and test the API of the backend application. An API is the way the frontend and backend talk to each other. Swagger creates an interactive web page where it is possible to see all the available API routes and send test requests directly from the browser, making it easy to check that the backend is working correctly.

FIGURE 2.7 – Swagger Logo

2.4.4 Technologies Used

React JS : React JS is a free and open-source JavaScript library created by Facebook. It is used to build the user interface of the web application. React makes it easy to create interactive pages by dividing the screen into small, reusable parts called components. Each component is responsible for one part of the page, which makes the code cleaner, easier to read, and easier to update.

FIGURE 2.8 – React JS Logo

JavaScript : JavaScript is a programming language that runs directly in the web browser. It is used to make web pages interactive and dynamic. In this project, JavaScript is the main language used for the frontend, since React JS is built on top of it. It allows the application to respond to user actions like clicking buttons, filling forms, and loading new data without refreshing the whole page.

FIGURE 2.9 – JavaScript Logo

CSS3 : CSS3, which stands for Cascading Style Sheets version 3, is a language used to control how web pages look. It sets colors, fonts, sizes, spacing, and the layout of elements on the screen. CSS3 also includes tools for making the interface responsive, which means the page automatically adjusts its layout to look good on different screen sizes, such as phones, tablets, and computers.

FIGURE 2.10 – CSS3 Logo

Vite : Vite is a modern build tool used during the development of the frontend. A build tool helps by automatically turning the source code into a format that the browser can understand and run. Vite is known for being very fast, which means the browser refreshes almost instantly every time a change is made to the code. This makes the development process much faster and more comfortable.

FIGURE 2.11 – Vite Logo

Spring Boot : Spring Boot is a Java framework used to build the backend of the application. A framework is a set of ready-made tools and rules that make it easier to develop software. Spring Boot makes it simple to create backend services that receive requests from the frontend, apply the business logic, and communicate with the database. It is widely used in professional projects because it is reliable, fast to set up, and easy to extend.

FIGURE 2.12 – Spring Boot Logo

JDK : The JDK, or Java Development Kit, is a set of tools needed to write, compile, and run Java programs. Compiling means turning the code written by the developer into a format that the computer can run. The JDK includes everything needed to develop Java applications, and it is required to use Spring Boot on the computer.

FIGURE 2.13 – JDK Logo

Python : Python is a simple and easy-to-read programming language that is widely used for artificial intelligence and data processing tasks. In this project, Python is used for the AI module, which is responsible for reading and analyzing CVs. It extracts useful information from uploaded CV files and calculates a score based on how well the candidate matches the job requirements.

FIGURE 2.14 – Python Logo

Node.js : Node.js is a tool that allows JavaScript code to run outside of the web browser, directly on the computer or server. Normally, JavaScript only works inside a browser, but with Node.js it can also be used for other tasks. In this project, Node.js is mainly used as part of the frontend development environment, since tools like npm, which is used to install JavaScript packages, need it to work.

FIGURE 2.15 – Node.js Logo

PostgreSQL : PostgreSQL is a free and open-source database management system used to store all the data of the application. A database is like a large and organized storage space where all information, such as user accounts, job offers, and candidate applications, is saved. PostgreSQL is reliable and supports many types of data operations, making it a good choice for web applications that need to store and manage large amounts of data.

FIGURE 2.16 – PostgreSQL Logo

2.5 Requirements Modeling

Requirements modeling provides a structured and visual representation of the system's expected behavior and the interactions between its different actors. In this project, the Unified Modeling Language (UML) is used to describe these interactions through use case diagrams, which serve as the primary tool for capturing and communicating functional requirements at a high level of abstraction.

2.5.1 Global Use Case Diagram

Use case diagrams are used to show the expected behavior of a system from the users' point of view. They help identify and organize the functional requirements by showing the interactions between users (actors) and the different system features. The system objectives must be clearly defined, and its development should be guided by the users' needs.

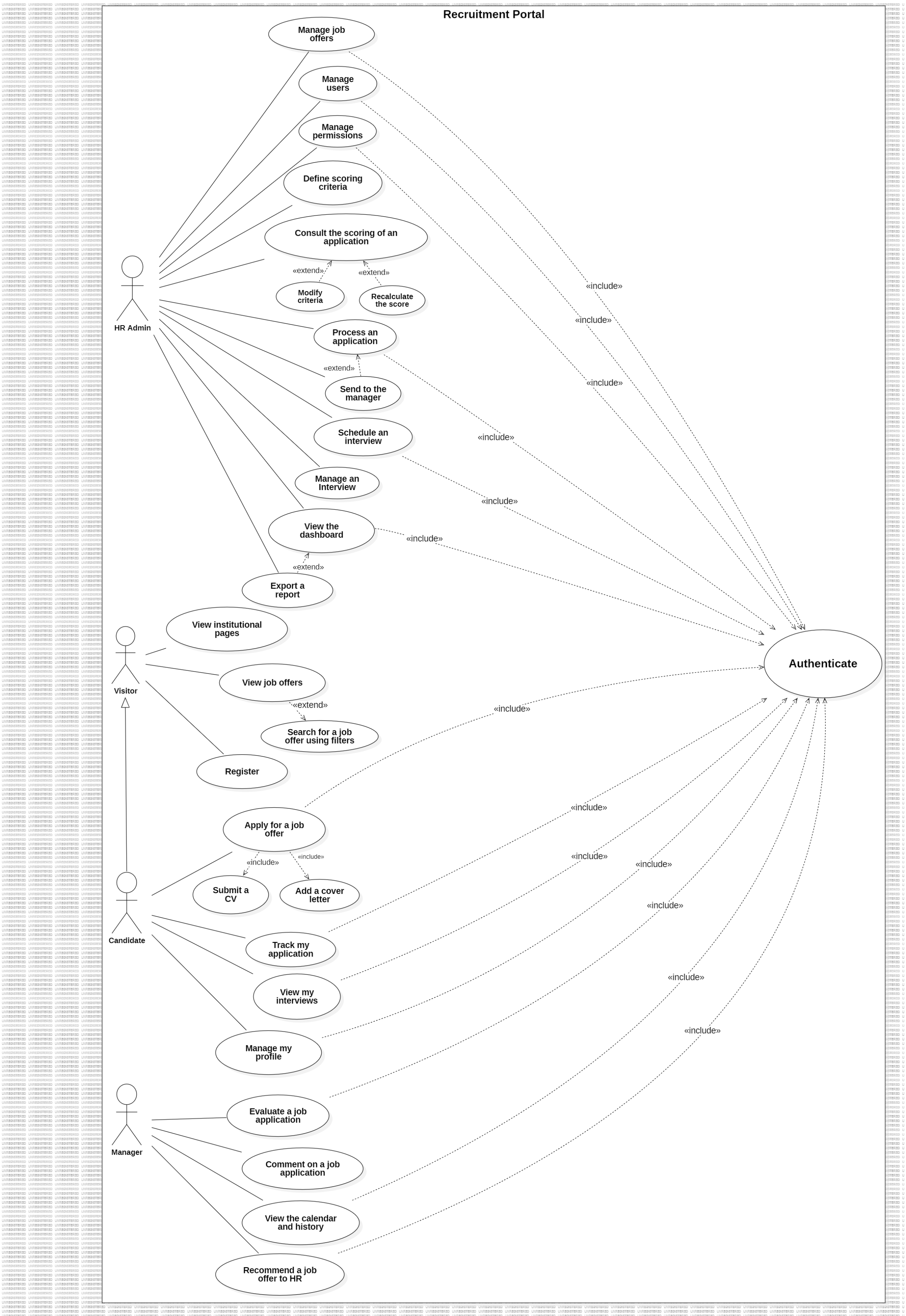

FIGURE 2.17 – Global Use Case Diagram

Conclusion

This chapter explained the preparation phase of the intelligent recruitment portal project. First, we identified the system users and defined both functional and non-functional requirements. Then, we described the SCRUM method used to manage the project, including team roles, the product backlog, and sprint planning. We also presented the working environment, including the hardware, software, and tools used during development. Finally, we showed the global use case diagram to give a clear overview of how users interact with the system. This preparation provides a strong base for the design and implementation chapters that come next.